

Medication Possession Ratio

2026-1-11

Data Wrangling and Cleaning:

The Group 1 and Group 2 datasets are two csv files that were originally created with the purpose of acting as a cumulative itemized bill for insurance members, listing unique medications, prescription durations, and transaction histories. Serving as subsets of BeniComp Health Solution's Google BigQuery Data Repository, the datasets were initially comprised of 2033 and 727 rows, as well as 7 variable columns: Member_ID, Claim_Number_and_Line, Date_Of_Service, Rx_Days_Prescribed, Rx_Name, NDC_Code, and Total_Paid_Sum. Out of these columns, five were character-type, one was integer-type, and one was double-type.

```
Group1 <- read.csv("C:/Users/smith/OneDrive/Desktop/Group1.csv")
Group2 <- read.csv("C:/Users/smith/OneDrive/Desktop/SQL csv's for med
project/Group2.csv")
print(head(Group1))
```

```
## Member.ID Claim.Number.and.Line Date.Of.Service Rx.Name
## 1 BCKVST-00 202108290004-01 2021-08-29 PFIZER COVID-19 VACC
## 2 BCKVST-00 202108300628-01 2021-08-29 PFIZER COVID-19 VACC
## 3 BCKVST-00 202108230815-01 2021-08-23 ARIPIPRAZOLE 5 MG
## 4 BCKVST-00 202108230816-01 2021-08-23 VENLAFAXINE HCL ER 7
## 5 BCKVST-00 202108080001-01 2021-08-08 PFIZER COVID-19 VA
## 6 BCKVST-00 202108040059-01 2021-08-04 HYDROCORTISONE 2.5%
## Rx.Days.Prescribed NDC.Code Total.Paid.Sum
## 1 1 59267100003 $40
## 2 1 59267100003 -$40
## 3 30 43598096630 $17
## 4 30 93738598 $9
## 5 1 59267100002 $40
## 6 30 51672300302 $0
```

```
print(head(Group2))
```

```
## Member_ID Claim_Number_and_Line Date_Of_Service Rx_Days_Prescribed
## 1 BLMVXD-00 202401311304-01 1/20/2024 30
## 2 BLMVXD-00 202401311115-01 1/16/2024 10
## 3 BLMVXD-01 202401311114-01 1/16/2024 30
## 4 BLMVXD-01 202402141200-01 2/12/2024 30
## 5 BLMVXD-01 202401160710-01 1/16/2024 15
## 6 BLMVXD-01 202401311113-01 1/16/2024 15
## Next_Fill_Date Days_Overlapped Days_Covered Rx_Name
## 1 2/19/2024 0 30 TADALAFIL
## 2 1/26/2024 0 10 CLINDAMYCIN HCL
## 3 2/15/2024 0 30 ESCITALOPRAM 20 MG T
## 4 3/13/2024 3 27 ESCITALOPRAM 20 MG T
```

## 5	1/31/2024	0	15	ALPRAZOLAM
## 6	1/31/2024	0	15	ALPRAZOLAM
##	Rx_Product_Code	NDC_Code	Total_Paid_Sum	MPR
## 1	114	27241011403	\$4	0.6666667
## 2	186	65862018601	\$25	0.2222222
## 3	282	43547028210	\$1	0.6666667
## 4	282	43547028210	\$3	0.6000000
## 5	1061	781106110	\$0	0.3333333
## 6	1061	781106110	\$0	0.3333333

```
print(str(Group1))
```

```
## 'data.frame': 2033 obs. of 7 variables:
## $ Member.ID : chr "BCKVST-00" "BCKVST-00" "BCKVST-00"
## $ Claim.Number.and.Line: chr "202108290004-01" "202108300628-01"
## $ Date.Of.Service : chr "2021-08-29" "2021-08-29" "2021-08-23"
## $ Rx.Name : chr "PFIZER COVID-19 VACC" "PFIZER COVID-19
VACC" "ARIPIRAZOLE 5 MG" "VENLAFAXINE HCL ER 7" ...
## $ Rx.Days.Prescribed : int 1 1 30 30 1 30 90 30 30 5 ...
## $ NDC.Code : num 5.93e+10 5.93e+10 4.36e+10 9.37e+07
5.93e+10 ...
## $ Total.Paid.Sum : chr "$40" "-$40" "$17" "$9" ...
## NULL
```

```
print(str(Group2))
```

```
## 'data.frame': 542 obs. of 12 variables:
## $ Member_ID : chr "BLMVXD-00" "BLMVXD-00" "BLMVXD-01"
## $ Claim_Number_and_Line: chr "202401311304-01" "202401311115-01"
## $ Date_Of_Service : chr "1/20/2024" "1/16/2024" "1/16/2024"
## $ Rx_Days_Prescribed : int 30 10 30 30 15 15 10 14 28 28 ...
## $ Next_Fill_Date : chr "2/19/2024" "1/26/2024" "2/15/2024"
## $ Days_Overlapped : int 0 0 0 3 0 0 0 0 0 0 ...
## $ Days_Covered : int 30 10 30 27 15 15 10 14 28 28 ...
## $ Rx_Name : chr "TADALAFIL" "CLINDAMYCIN HCL" "ESCITALOPRAM
20 MG T" "ESCITALOPRAM 20 MG T" ...
## $ Rx_Product_Code : int 114 186 282 282 1061 1061 840 441 342 342
...
## $ NDC_Code : num 2.72e+10 6.59e+10 4.35e+10 4.35e+10
7.81e+08 ...
## $ Total_Paid_Sum : chr "$4 " "$25 " "$1 " "$3 " ...
## $ MPR : num 0.667 0.222 0.667 0.6 0.333 ...
## NULL
```

```
print(is.null(Group1$Total.Paid.Sum))

## [1] FALSE

print(is.null(Group2$Total_Paid_Sum))

## [1] FALSE

print(summary(Group1))

##      Member.ID          Claim.Number.and.Line Date.Of.Service      Rx.Name
## Length:2033          Length:2033          Length:2033      Length:2033
## Class :character     Class :character     Class :character     Class
## Mode  :character     Mode  :character     Mode  :character     Mode
## Mode  :character
##
##
##
##
## Rx.Days.Prescribed      NDC.Code          Total.Paid.Sum
## Min.   : 0.00          Min.   :0.000e+00      Length:2033
## 1st Qu.: 15.00          1st Qu.:6.427e+08      Class :character
## Median : 30.00          Median :5.074e+10      Mode  :character
## Mean   : 46.84          Mean   :3.953e+10
## 3rd Qu.: 90.00          3rd Qu.:6.534e+10
## Max.   :900.00          Max.   :8.770e+10
## NA's   :1              NA's   :5

print(summary(Group2))

##      Member_ID          Claim_Number_and_Line Date_Of_Service
## Rx_Days_Prescribed
## Length:542          Length:542          Length:542      Min.   : 1.00
## Class :character     Class :character     Class :character  1st Qu.:10.00
## Mode  :character     Mode  :character     Mode  :character  Median :30.00
##                                     Mean   :37.58
##                                     3rd Qu.:60.00
##                                     Max.   :90.00
##
## Next_Fill_Date        Days_Overlapped      Days_Covered      Rx_Name
## Length:542          Min.   : 0.0000      Min.   : 0.00      Length:542
## Class :character     1st Qu.: 0.0000      1st Qu.:10.00      Class :character
## Mode  :character     Median : 0.0000      Median :30.00      Mode  :character
##                                     Mean   : 0.7362      Mean   :36.84
##                                     3rd Qu.: 0.0000      3rd Qu.:56.00
##                                     Max.   :76.0000      Max.   :90.00
##
## Rx_Product_Code        NDC_Code          Total_Paid_Sum      MPR
## Min.   : 1.0          Min.   :2.151e+06      Length:542          Min.   :0.0000
## 1st Qu.: 186.2        1st Qu.:8.437e+09      Class :character     1st Qu.:0.2222
## Median : 526.0        Median :5.186e+10      Mode  :character     Median :0.6667
## Mean   :1448.5        Mean   :4.123e+10      Mean   :0.8187
```

```
## 3rd Qu.:1183.0 3rd Qu.:6.586e+10 3rd Qu.:1.2444
## Max. :9887.0 Max. :8.078e+10 Max. :2.0000
```

```
Group1_2021 <- read.csv("C:/Users/smith/OneDrive/Desktop/SQL csv's for med
project/Group1(2021).csv")
Group1_2022 <- read.csv("C:/Users/smith/OneDrive/Desktop/SQL csv's for med
project/Group1(2022).csv")
Group1_2023 <- read.csv("C:/Users/smith/OneDrive/Desktop/SQL csv's for med
project/Group1(2023).csv")
Group1_2024 <- read.csv("C:/Users/smith/OneDrive/Desktop/SQL csv's for med
project/Group1(2024).csv")
Group2 <- read.csv("C:/Users/smith/OneDrive/Desktop/SQL csv's for med
project/Group2.csv")
print(summary(Group2))
```

```
## Member_ID Claim_Number_and_Line Date_Of_Service
Rx_Days_Prescribed
## Length:542 Length:542 Length:542 Min. : 1.00
## Class :character Class :character Class :character 1st Qu.:10.00
## Mode :character Mode :character Mode :character Median :30.00
## Mean :37.58
## 3rd Qu.:60.00
## Max. :90.00
## Next_Fill_Date Days_Overlapped Days_Covered Rx_Name
## Length:542 Min. : 0.0000 Min. : 0.00 Length:542
## Class :character 1st Qu.: 0.0000 1st Qu.:10.00 Class :character
## Mode :character Median : 0.0000 Median :30.00 Mode :character
## Mean : 0.7362 Mean :36.84
## 3rd Qu.: 0.0000 3rd Qu.:56.00
## Max. :76.0000 Max. :90.00
## Rx_Product_Code NDC_Code Total_Paid_Sum MPR
## Min. : 1.0 Min. :2.151e+06 Length:542 Min. :0.0000
## 1st Qu.:186.2 1st Qu.:8.437e+09 Class :character 1st Qu.:0.2222
## Median :526.0 Median :5.186e+10 Mode :character Median :0.6667
## Mean :1448.5 Mean :4.123e+10 Mean :0.8187
## 3rd Qu.:1183.0 3rd Qu.:6.586e+10 3rd Qu.:1.2444
## Max. :9887.0 Max. :8.078e+10 Max. :2.0000
```

#Convert the data type of the "Rx_Product_Code" within the Group 2 data table

```
r = getOption("repos")
r["CRAN"] = "http://cran.us.r-project.org"
options(repos = r)
install.packages("data.table")
```

```
## Installing package into 'C:/Users/smith/AppData/Local/R/win-library/4.3'
## (as 'lib' is unspecified)
```

```
##
## There is a binary version available but the source version is later:
## binary source needs_compilation
## data.table 1.17.0 1.18.0 TRUE
```

```

##
## Binaries will be installed
## package 'data.table' successfully unpacked and MD5 sums checked
##
## The downloaded binary packages are in
## C:\Users\smith\AppData\Local\Temp\Rtmp48A8j6\downloaded_packages

library("data.table")

## Warning: package 'data.table' was built under R version 4.3.3

sapply(Group2,class)

##           Member_ID Claim_Number_and_Line      Date_Of_Service
##           "character"           "character"           "character"
## Rx_Days_Prescribed      Next_Fill_Date      Days_Overlapped
##           "integer"           "character"           "integer"
##           Days_Covered           Rx_Name      Rx_Product_Code
##           "integer"           "character"           "integer"
##           NDC_Code      Total_Paid_Sum           MPR
##           "numeric"           "character"           "numeric"

setDT(Group2)
Group2_new <- Group2[ , "Rx_Product_Code" := as.character("Rx_Product_Code")]

## Warning in `[.data.table`(Group2, , `:=`("Rx_Product_Code",
## as.character("Rx_Product_Code"))): Coercing 'character' RHS to 'integer'
## to
## match the type of column 9 named 'Rx_Product_Code'.

## Warning in `[.data.table`(Group2, , `:=`("Rx_Product_Code",
## as.character("Rx_Product_Code"))): NAs introduced by coercion

sapply(Group2_new,class)

##           Member_ID Claim_Number_and_Line      Date_Of_Service
##           "character"           "character"           "character"
## Rx_Days_Prescribed      Next_Fill_Date      Days_Overlapped
##           "integer"           "character"           "integer"
##           Days_Covered           Rx_Name      Rx_Product_Code
##           "integer"           "character"           "integer"
##           NDC_Code      Total_Paid_Sum           MPR
##           "numeric"           "character"           "numeric"

summary(Group2_new)

##   Member_ID      Claim_Number_and_Line Date_Of_Service
## Rx_Days_Prescribed
## Length:542      Length:542      Length:542      Min.    : 1.00
## Class :character Class :character Class :character 1st Qu.:10.00
## Mode  :character Mode  :character Mode  :character Median :30.00
##                                     Mean  :37.58

```

```

##                                     3rd Qu.:60.00
##                                     Max.    :90.00
##
## Next_Fill_Date      Days_Overlapped    Days_Covered    Rx_Name
## Length:542          Min.    : 0.0000    Min.    : 0.00    Length:542
## Class :character    1st Qu.: 0.0000    1st Qu.:10.00    Class :character
## Mode  :character    Median : 0.0000    Median :30.00    Mode  :character
##                    Mean   : 0.7362    Mean   :36.84
##                    3rd Qu.: 0.0000    3rd Qu.:56.00
##                    Max.    :76.0000    Max.    :90.00
##
## Rx_Product_Code      NDC_Code          Total_Paid_Sum    MPR
## Min.    : NA          Min.    :2.151e+06    Length:542        Min.    :0.0000
## 1st Qu.: NA          1st Qu.:8.437e+09    Class :character    1st Qu.:0.2222
## Median : NA          Median :5.186e+10    Mode  :character    Median :0.6667
## Mean   :NaN          Mean   :4.123e+10                    Mean   :0.8187
## 3rd Qu.: NA          3rd Qu.:6.586e+10                    3rd Qu.:1.2444
## Max.    : NA          Max.    :8.078e+10                    Max.    :2.0000
## NA's    :542

```

SQL QUERIES WITH EXPLORATORY QUESTIONS (Group 1 queries will display formatting for each question; outputs for each group is displayed below):

1) Top 5 medications prescribed?

```
1 • SELECT NDC_Code, count(NDC_Code) AS CodeCount FROM `group1(2021)`.`group1(2021)`
2   Group BY NDC_Code
3   Order BY CodeCount DESC
4   Limit 5;
```

NDC_Code	CodeCount	NDC_Code	CodeCount	NDC_Code	CodeCount	NDC_Code	CodeCount
65862000899	4	57664049918	3	59746000103	2	74706819	3
27808003502	4	16729048601	3	69584085290	2	3089421	2
61958250101	3	62756035664	2	68180035202	2	603459315	2
16714043901	3	00642747001	2	72618300002	2	16729022517	2
00093738598	2	66993058597	2	64380079901	2	75834022101	2

G1_2021

G1_2022

G1_2023

G1_2024

NDC_Code	CodeCount
62332025118	6
00093317431	6
64980034214	5
65862044990	5
66993007996	5

G2

2)Top 5 patients with the most prescriptions

```
1 • SELECT Member_ID, count(NDC_Code) AS CodeCount FROM `group1(2021)`.`group1(2021)`
2   Group BY Member_ID
3   Order BY CodeCount DESC
4   Limit 5;
```

Member_ID	CodeCount	Member_ID	CodeCount	Member_ID	CodeCount	Member_ID	CodeCount
HRVLGR-00	13	CGSRHJ-00	7	CGSRHJ-00	5	GWZHDK-00	8
VCKLZH-00	13	XJLXFG-00	7	GFNBH-00	5	FMTXFR-00	6
BCKVST-00	8	GWZHDK-00	5	XJLXFG-00	5	GCJHGN-01	5
JXXVRX-00	7	JFVVZG-00	4	GWZHDK-00	4	HHKPHH-00	5
VKPBCS-00	6	PJJHBK-00	3	MFQPSX-00	4	VKPBCS-00	5

G1_2021

G1_2022

G1_2023

G1_2024

	Member_ID	CodeCount
▶	BWTHVD-01	20
	VRZXND-01	14
	QHNVQL-01	13
	DHXVXH-01	12
	FVXJBD-12	11

G2

3)Top five prescription durations grouped by medication?

```

1 • SELECT
2     NDC_Code, SUM(Days_Covered) AS Total_Days_Covered FROM `group1(2021)`.`group1(2021)`
3 GROUP BY
4     NDC_Code
5 ORDER BY
6     Total_Days_Covered DESC
7 LIMIT 5;

```

	NDC_Code	Total_Days_Covered
▶	65862000899	270
	00093738598	90
	64980043910	90
	68382000914	90
	67877019710	90

	NDC_Code	Total_Days_Covered
▶	57664049918	182
	31722053705	180
	16729048601	168
	63304082805	90
	16729011917	90

	NDC_Code	Total_Days_Covered
▶	52817035910	180
	69584085290	180
	68180035202	180
	65862022801	90
	31722072690	90

	NDC_Code	Total_Days_Covered
▶	16729022517	180
	59651021430	100
	378718505	90
	43598083005	90
	65862037405	90

G1_2021

G1_2022

G1_2023

G1_2024

	NDC_Code	Total_Days_Covered
▶	65162080710	357
	42806054701	264
	66993007996	240
	70010020510	210
	55111072590	194

G2

```
Ratios <- read.csv("C:/Users/smith/OneDrive/Desktop/SQL csv's for med
project/Ratios.csv")
head(Ratios)
```

```
##           MPR    Group
## 1 0.0000000 G1_2021
## 2 2.0000000 G1_2021
## 3 0.0000000 G1_2021
## 4 0.6666667 G1_2021
## 5 0.0000000 G1_2021
## 6 0.6666667 G1_2021
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.3.2
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:data.table':
```

```
##
```

```
##      between, first, last
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##      filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      intersect, setdiff, setequal, union
```

```
group_by(Ratios, Group) %>%
  summarise(
    count = n(),
    mean = mean(MPR, na.rm = TRUE),
    SD = sd(MPR, na.rm = TRUE),
    median = median(MPR, na.rm = TRUE),
    IQR = IQR(MPR, na.rm = TRUE)
  )
```

```
## # A tibble: 5 × 6
```

```
##   Group    count  mean    SD median  IQR
##   <chr>   <int> <dbl> <dbl> <dbl> <dbl>
## 1 G1_2021     86 0.438 0.656 0.0111 0.667
## 2 G1_2022     38 1.16  0.727 1      1.33
## 3 G1_2023     67 0.898 0.774 0.667  1.81
## 4 G1_2024     56 1.06  0.789 0.667  1.47
## 5 G2         295 0.808 0.670 0.667  0.344
```

```
install.packages("ggpubr")
```

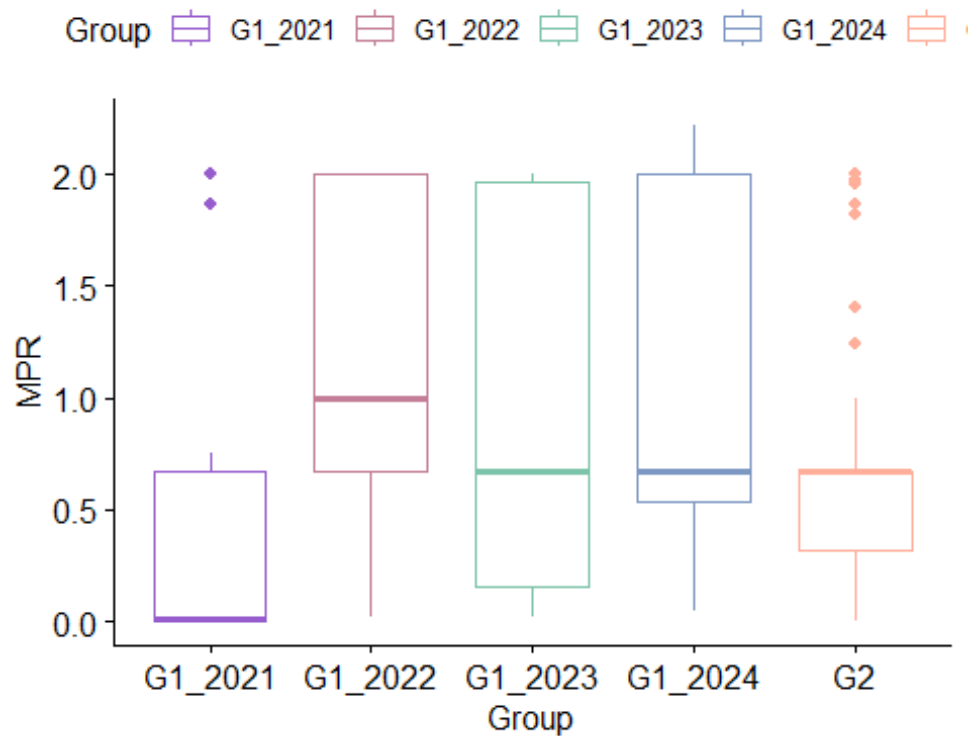
```
## Installing package into 'C:/Users/smith/AppData/Local/R/win-library/4.3'
## (as 'lib' is unspecified)

##
##   There is a binary version available but the source version is later:
##       binary source needs_compilation
## ggpubr 0.6.0 0.6.2 FALSE
## installing the source package 'ggpubr'

library(ggpubr)

## Loading required package: ggplot2

ggboxplot(Ratios, x= "Group", y= "MPR", color = "Group", palette =
c("#975ccd", "#c37d97", "#7dc3a9", "#7d97c3", "#ffae99"), order = c("G1_2021",
"G1_2022", "G1_2023", "G1_2024", "G2"), ylab = "MPR", xlab = "Group")
```



When interpreting the boxplot outputs, the clear trends stemmed the behavior of G1_2021 and G2's data. Although the other three groups (G1_2022, G1_2023, G1_2024) had data plotted in similar areas, as seen by each median being within the span of another group's data spread, the interquartile ranges (IQR) of G1_2021 and G2 were well outside of the other groups'. Furthermore, these patient groups contained outliers in which the medication possession ratios were closer to 2.0. In these instances, patients were received extensive prescriptions last upwards of 3 months. For G1_2021 in particular, several BeniComp insurance holders received prescriptions for Venlafaxine, a depression-treating drug that was ranking the 44th most popular prescription in 2021. Due to the negative psychological and socioeconomic impacts of the COVID-19 pandemic, the uncertainty as to when the pandemic would end most likely influenced an extended dependence on antidepressants.

```
kruskal.test(MPR ~ Group, data = Ratios)

##
##  Kruskal-Wallis rank sum test
##
## data:  MPR by Group
## Kruskal-Wallis chi-squared = 62.618, df = 4, p-value = 8.165e-13

pairwise.wilcox.test(Ratios$MPR,Ratios$Group,p.adjust.method = "none")

##
##  Pairwise comparisons using Wilcoxon rank sum test with continuity
##  correction
##
## data:  Ratios$MPR and Ratios$Group
##
##           G1_2021 G1_2022 G1_2023 G1_2024
## G1_2022 2.6e-08 -         -         -
## G1_2023 5.1e-07 0.0393 -         -
## G1_2024 6.1e-09 0.5990 0.0622 -
## G2      8.8e-11 0.0025 0.8949 0.0208
##
## P value adjustment method: none
```

To assess whether there were any statistically significant differences between the medication possession ratios of each group, a Kruskal-Wallis test was performed with a significance level of .05. The null hypothesis is that the MPR median for each group is the same, whereas the alternative hypothesis is that the medians differ. Recalling the chosen significance level (.05) and utilizing the test output of p-value= 8.165e-13, the p-value being less than 0.05 concludes that we reject the null hypothesis and assume that there are differences between the patient groups.

Futhermore, although the findings from the Kruskal-Wallis test were utilized to identify any important statistical differences, the Pairwise Wilcoxon Test was implemented with a p-value of 0.05 to identify the patient group pairings in which this differences exist and

determine whether the patients come from the same population. As seen in the test results, the following Group 1 and Group 2 pairings were less than 0.05:(G1_2021 and G2), (G1_2022 and G2), and (G1_2024 and G2). Therefore, we can reject the null hypothesis and conclude that there is statistically significant difference between the groups. Regarding G1_2023 being above 0.05 and G1_2024 being the closest of the pairings that were under the significance level, there are less statistically significant differences, and the results can likely be attributed to G1_2023, G1_2024, and G2 having data from similar timeframes. Similarly, by identifying G1_2021 pairings as outputting more values below 0.05, as well as referring to first SQL query question that investigated the most prescribed medications, we can specifically determine how health trends in 2021 impacted medication adherence and prominence. Within Group 1's 2021 data, the most prescribed medications were Metformin and Hydrocodone. Metformin has been commonly utilized to control high blood sugar levels in carriers of Type 2 Diabetes, which consisted of roughly 11.6% of Americans in 2021. On the other hand, the National Library of Medicine recognized Hydrocodone's influence when treating non-productive cough, a symptom of COVID-19.